

A Play-by-Play Analysis of Attendance Effects in College Football

by

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Author's Declaration

I hereby declare that I am the sole author of this Master's Research Paper (MRP). This is a true copy of the MRP, including any required final revisions, as accepted by my examiners.

Abstract

We provide the first play-by-play analysis of the effect of attendance on outcomes and penalties in college football. Using cfbfastR data on 7,487 games from 2014 to 2024 we exploit the heterogeneity in collegiate attendance that the NFL literature lacks. We examine three outcomes in the form of play results, penalties, and early timeouts. We find that each additional 10,000 fans costs the visiting offense 0.032 yards and 0.005 EPA per play. The home offense shows no significant positive effect and thus the crowd effect operates by degrading the visiting offense rather than by helping the home offense. Away offenses commit more false starts as the crowd grows while home defenses jump offside less. Away offenses also call more early timeouts while defensive timeouts show no effect. These effects grow as the down increases as crowds are loudest on third downs. Taken together our results point to crowd noise physically degrading the visiting offense's pre-snap communication. Our results are robust to a variety of specifications and controls such as an IV model using the 2020 COVID season similar to Cook (2026).

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List of Abbreviations

2SLS	Two-stage least squares
Def	Defensive
EPA	Expected points added
Elo	Elo rating (team strength)
FBS	Football Bowl Subdivision
FE	Fixed effects
FS	False start
GBG	Game-by-game
H2H	Head-to-head
HFA	Home field advantage
IV	Instrumental variable
LPM	Linear probability model
MRP	Master's Research Paper
NCAA	National Collegiate Athletic Association
NFL	National Football League
Off	Offensive
OLS	Ordinary least squares
OLS nC	OLS excluding the 2020 COVID season
OLS nF	OLS without season fixed effects
OVb	Omitted variable bias
PBP	Play-by-play
PI	Pass interference

1 Introduction

1.1 Background

In NCAA collegiate football among FBS teams the home team wins 58.4% of their games (see Appendix B, Table 14). Part of this effect is attributed to the home team being more familiar with their stadium and not needing to travel before games. However, research has pointed out that another portion of this effect comes from the home crowd (Cook, 2026; Farnell, 2023; McMahon and Quintanar, 2024). The home crowd can affect the game in two ways. First, with physical noise. For example, if teammates cannot communicate the planned play to one another mishaps occur. The ball may fail to be snapped, the team may take an accidental pre-snap penalty, or a receiver may run the wrong route. This is unique to American football when compared to other sports such as basketball, baseball, or hockey. Second, the home crowd can influence the mental states of individuals involved with the games. This can range from coaches to players to referees. The issue with separating these effects is that data on the decibel level of noise is not available. Thus we must rely on attendance counts and percentages to test different channels to unveil the mechanism.

1.2 Gaps and Research Questions

Recent work has arisen due to the natural experiment of COVID in 2020 which limited how many fans were allowed in stadiums. However most has been done at the NFL level (Cook, Farnell), where attendance lacks heterogeneity. Figure 1 demonstrates the vast difference in variation between the two leagues. The right panels plot weekly attendances as totals and percentages for every NCAA FBS football team in 2022. The left panels plot the same values but for the NFL. The plots of the raw attendance counts show how college football has a much wider range of attendance values than the NFL. The percentage plots then show that fan counts barely move at the team level within a season in the NFL.

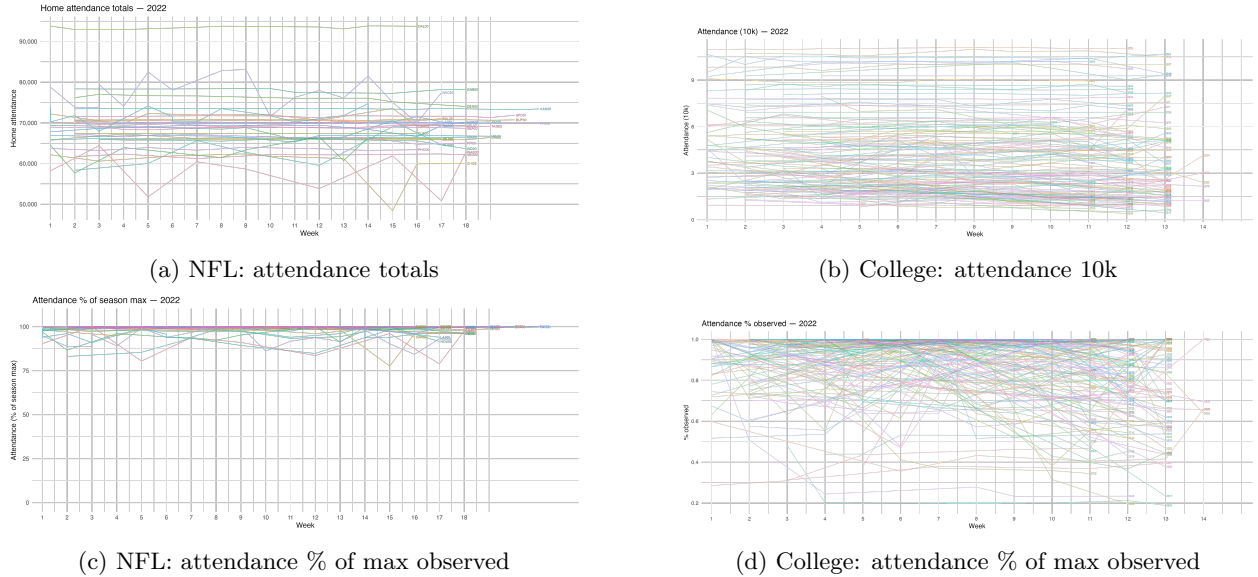


Figure 1: Attendance variation: NFL vs college

This natural variation in college football drives our identification. NFL attendance is stable outside of 2020. On the other hand, college attendance varies within teams and within season. The effect therefore is identified off of two additional sources of variation in college football rather than solely the COVID shock. We have team-by-team variation which comes from there being smaller market teams in this sample. Variation can also occur within a team across weeks as college crowds react more to their teams going on winning or losing streaks than crowds in the NFL.

Work in college football uses at game level data (McMahon and Quintanar, 2024) but lacks analysis of penalties which is a key tool to pin down the mechanism. Given this, there is a large gap in the literature that is missing play-by-play data that contains information about penalties at the collegiate level. In filling this gap we are able to answer the following research questions.

- Does home attendance causally influence the outcome of college games?
- By what mechanism does attendance affect game outcomes? Is it referee bias, noise, influenced play calling or any other un-discussed variables?
- Does heterogeneity in attendance lead to a difference in results when compared to NFL literature? Furthermore, is the lack of heterogeneity in NFL attendance the reason that Cook (2026) does not find concrete evidence of endogeneity?

1.3 Contribution and Motivation

We contribute the first play-by-play analysis of the effect of attendance on outcomes and penalties at the collegiate level. Our improved heterogeneity also helps us identify significant effects at the play-by-play level beyond what Farnell (2023) could identify.

These contributions are needed as football is one of the most popular sports in America. It is played at an enormous scale across the professional, collegiate, and high school levels (Benz et al., 2024). Teams are incentivized to win, as winning increases demand for a team’s product (Borland & Macdonald, 2003). Thus if attendance is a lever that can be pulled to influence the outcome of a game teams should consider it. For example if lowering ticket prices below the short-term profit maximizing level can increase attendance and thus increase the probability of winning then it may be in the teams best interest to do so.

Penalties are what let us separate referee bias from crowd noise. Some penalties, such as false starts, leave little to no room for referee judgment and are thus non-discretionary. However a penalty such as pass interference is a discretionary call that can be biased by the referee. If larger crowds influence only the discretionary penalties then one can argue that the mechanism is through referee bias. However if they instead move only non-discretionary penalties then one can argue that the mechanism works through noise. It is these differences across penalty types that let us pin down whether the mechanism comes from referee bias or not. In the case that referee bias is affecting games, leagues have a governance problem that must be resolved for the integrity of their game. If the mechanism is the actual noise of the crowd then offenses and defenses need to adjust their play calling accordingly to accommodate this.

1.4 Roadmap

The remainder of this paper is organized as follows. Section 2 reviews relevant literature. Section 3 describes our data. Section 4 presents the methodology. Section 5 reports our results. Section 6 concludes our paper.

2 Literature Review

2.1 Crowd Noise

Work on the home team effect on the outcome of sports games began with Schwartz & Barsky (1977) in which they use 1971 data on baseball, football, hockey, basketball to show that the home field advantage exists. They also show that this effect increases when games are held indoors which suggests some form of noise channel.

Losak & Sabel (2021) use MLB data from the 2019 and 2020 seasons. They find that the absence of a crowd does not reduce the home field advantage. This can be attributed to how baseball has the least verbal communication among popular sports.

An opportunity for a natural experiment arose in which Italian soccer stadiums were severely reduced in terms of capacity due to security measures following the hooligan incident in Sicily on February 2, 2007. Taking advantage of this Pettersson-Lidbom & Priks (2010) find a significant effect where home teams are favoured more in games with crowds than without crowds. In soccer, Cross & Uhrig (2023) use Covid as a natural experiment to show that the home team advantage is reduced by over 50% when crowds are absent. Similarly Scoppa (2021) argues for a psychological effect in European soccer leagues using a difference-in-difference approach. He finds that the HFA nearly halves in terms of points, goals and shots.

Ehrlich & Potter (2023) use COVID as a natural experiment and uncover that when there are no crowds present the HFA entirely vanishes. Interestingly though, they find that a sparse crowd is not statistically different from a full crowd.

Boudreaux, Sanders & Walia (2017) use the Lakers and Clippers who share Staples Center and played 59 games from 1999–2014 to pin down the effect of the crowd when both teams have the exact same level of familiarity with the stadium. In this they find that the “arbitrary” home-team designation significantly raises win probability by 21–22.8 percentage points which is shockingly similar to the entire NBA’s home team advantage of 20.4 percentage points. They attribute this effect to the crowd as season ticket holders let the crowd composition sway in favour of the home team.

2.2 Football Specific Controls

Craig et al. (2016) lay foundational work and show that a “non-football” variable such as the temperature can influence penalty aggression in the NFL. They find that as the temperature increases the home team commits more aggressive penalties. This affects us in two ways. First we must control for weather and second it should make us pause and note that the performance of players in the NFL, despite their professionalism, are affected by their physical senses. If we take a step further we ask if the sensation of noise can affect a player’s performance outside of miscommunications caused by that noise or referee bias.

Nichols (2014) found that travel distance and time zone changes affect the home team advantage as well. Away teams that travel longer distances and across more west-to-east time zones have worse away team performance. This aligns with the findings of Benz et al. (2024) who note a decline in home field advantage in the NFL/NCAA but not in High School football. They attribute this to adoption of instant replay and improvements to travel at high levels.

2.3 Referee Bias

Snyder and Lopez (2015) study discretionary penalties in the NFL with data from 2002 to 2012. They reveal that defensive pass interference is 20% more likely when the home team is on offense. Lopez (2016) shows that referees also succumb to sideline pressure. Penalties against a team are more likely when the play occurs near their opponent’s sideline, where coaches and players can pressure officials. Brymer et al. (2021) also show that referees bias towards the underdog team in collegiate football within same conference games. Bryson et al. (2021) find that in soccer fewer yellow cards are awarded to away teams during the Covid season when crowds are absent.

In hockey, the home team’s advantage shrinks when games are played in empty arenas using Covid as a natural experiment. Guérette et al. (2021) find NHL and CHL referees award more penalties to away teams when a crowd is present, while Thrane (2024) shows in Norwegian hockey that home teams receive more suspensions without a crowd, which in turn increases away team scoring.

Fischer & Haucap (2021) find no evidence of referee bias in the Bundesliga. Interestingly though, using covid, they find a drop in the home team advantage with restricted crowd only in the 1st division. In the 2nd and

3rd divisions they find no effect. They attribute this to lower division teams being mentally prepared and used to sparse attendance.

Ehrlich & Potter (2023) conclude that there is not significant effect of the crowds on the officiating of the game. Gong (2022) echoes this and finds no evidence of referee bias between home and away teams in the NBA.

Penalties in soccer, hockey and basketball speak well to referee behaviour. However the ball is not "snapped" in these sports and thus there is no pre-snap moment where noise can influence the player's communication. In American football penalties can measure both channels if they are split by type. Farnell (2023) splits them in the NFL. We are the first to do so in college.

2.4 Identification

Smith and Groetzinger (2010) never directly state a simultaneity bias. It is inferred by Cook (2026). They use MLB game level data from 1996-2005 to examine the effect of attendance, measured as a % of capacity, on a home team win indicator. Attendance is the endogenous regressor and is instrumented with the game's day of the week and the temperature at game time. They pitch a reverse causality story where teams that are expected to win more will have a higher attendance. Their Durbin-Wu-Hausman tests reject the null of exogeneity comfortably thus confirming some source of endogeneity that is being corrected for.

Cook (2026) infers this to be simultaneity and instruments attendance with COVID. His LPM and IV coefficients come out nearly identical. Reed (2015) shows that a lagged suspected variable may be used if it satisfies exclusion and relevance. We show in our methodology that this condition fails for us as the threat comes from potential omitted variables and not simultaneity.

COVID is the dominant source of variation in this literature. Cross and Uhrig (2023), Ehrlich and Potter (2023), Fischer and Haucap (2021), Bryson et al. (2021), and Guérette et al. (2021) all use it as a natural experiment. Scoppa (2021) uses a difference-in-differences design. Pettersson-Lidbom and Priks (2010) use the Italian capacity restrictions as a pre-COVID version of the same shock. Boudreaux et al. (2017) instead hold stadium familiarity fixed by using two teams that share an arena.

Ehrlich et al (2024) is the predecessor to Cook (2026). They find the presence of the home crowd itself, not crowd density, is responsible for the entirety of observed NFL home field advantage. Cook (2026) improves on them, noting they overlook referee bias, weather, and time-zone travel. Again we can point to the lack of heterogeneity in attendance in the NFL as a potential reason for their results.

McMahon and Quintanar (2024) use no instrument. They split their sample into pre 2020 and during 2020 and compare the two. These results directly improve upon an insignificant crowd effect from Wang et al. (2011) which we now know is due to sample size.

The majority of the above leans on the shock of COVID as the NFL and other professional sports have little variation in attendance. Figure 1 shows that college attendance does. Thus we are able to identify crowd effects off more than solely the shock of COVID.

2.5 Nearest Neighbors

Cook (2026) used game-by-game data from the 2009-2022 using the 2020 natural experiment of Covid as an IV. He regresses attendance and standard game controls on home win %, home point margin, and various total penalties measures. In this he finds a causal 2.11 point advantage at average attendance which accounts for 97% of the 2.17 home point advantage. He also notes that total penalties are higher for both home and away teams which suggests that referee bias is not the mechanism. Furthermore he shows that the interacted indoor effect is twice as large which again points to the mechanism being noise. It is also important to note that his LPM and IV coefficients are nearly identical. He pointed to simultaneity bias as a potential source of endogeneity (that he used a IV for). The citation he provides for this is Smith and Groetzinger (2010) which we discussed above.

Farnell (2023) uses play-by-play data from the 2018–2020 NFL seasons to examine the effect of attendance on penalties. Most importantly he splits his data by penalty type. Specifically he looks at “non-discretionary pre-snap penalties” which are penalties that are presnap, clear and free of referee bias. Since this is play-by-play data the regressions are run from the perspective of the defensive team with an interaction term of percent full x home team. He finds that home defenses commit fewer penalties at high attendance relative to away defenses. In his paper there is no concrete evidence on which team false starts more due to crowds. It is important to note that there is no IV within this paper.

McMahon and Quintanar (2024) use collegiate game-by-game data from 2009–2020. They run regressions from each team’s perspective and thus there are two observations per game. They also include neutral site games. They employ OLS and rely on 2020 as a shock to separate effects into pre and during 2020. They find that for every 38,875 fans the away team is estimated to score 1 less point. In a separate regression they find that when split pre 2020 and during 2020 the effect dies in the 2020 season. Since in 2020 college teams were allowed to pump in noise to a specified decibel cap they argue that the true mechanism is noise as the attendance in 2020 does not affect noise but rather now purely individuals-in-seats.

3 Data

3.1 Sources

All game-level outcomes, play-by-play data, venue information, and team identifiers are obtained through the cfbfastR package (Gilani et al., 2025), which provides access to the College Football Data API (collegefootballdata.com). Game time weather is drawn from the Open Meteo historical weather archive (Zippenfenig, 2023). It is then matched to each venue by latitude, longitude, and local kickoff time.

Our sample is the 2014 to 2024 seasons as this is when play-by-play data becomes available. We keep completed regular season games between FBS teams at non-neutral sites and drop any game with missing attendance. At the play level we keep rush and pass plays on downs one through three and drop kneels and spikes. The play-by-play file also drops plays wiped by a penalty which is why it is a subset of the penalty file.

To measure attendance we have two different variables. Our main specification is attendance (10k) which is simply the attendance in terms of 10,000 fans. The second variable is obtained by first storing the maximum observed attendance for a given stadium in a given season. If the season is Covid then we instead store the previous season’s max observed attendance. We then divide the attendance by this observed max to get the attendance (% obs).

Below are summary tables of our main penalty data set. Due to the similarity of the data please see Appendix B for play-by-play, game-by-game and timeout level data. Note that the play-by-play level data is a subset of the penalty play-by-play data as each observation in the play-by-play data is for a play that was run and had a result that was penalty free. Thus the dependent variables in this data will be if a first down was obtained and the yards gained on the play. The penalty play-by-play data includes all plays. Thus the dependent variables in this data will be if a penalty of a certain type was called. See appendix B for our game-by-game summary statistics.

In our play-level data we acknowledge a point spike in yards gained at 0 which is natural due to incomplete passes. These are not censored or missing values thus we estimate with OLS. Also note that we control for if the play is a pass play.

3.2 Tables

See appendix B for our game-by-game, play-by-play and timeout level summary statistics.

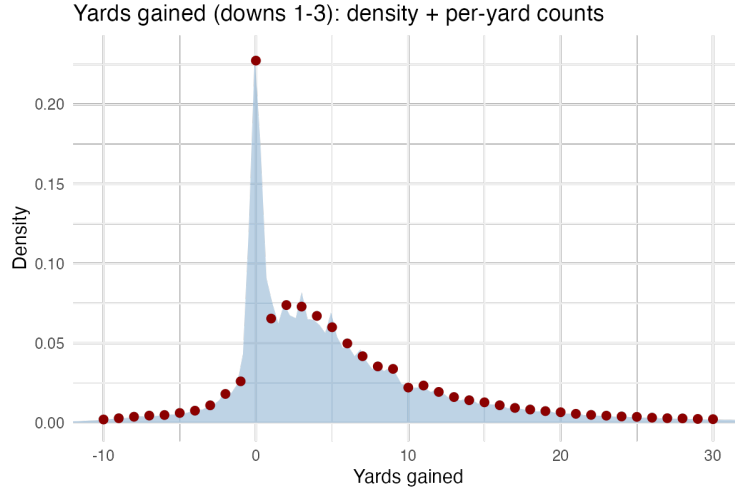


Figure 2: Distribution of yards gained on downs 1–3, with per-yard play counts (red) overlaid on the kernel density.

3.2.1 PBP Penalties

Table 1: Summary statistics: penalty-level model variables (downs 1–3)

Variable	N	NA	Mean	SD	Min	Max
Outcomes						
False start	1,058,548	0	0.016	0.124	0.000	1.000
Presnap off (ex FS)	1,058,548	0	0.004	0.062	0.000	1.000
Def offside	1,058,548	0	0.006	0.075	0.000	1.000
Presnap def (ex offside)	1,058,548	0	0.001	0.036	0.000	1.000
Off holding	1,058,548	0	0.012	0.108	0.000	1.000
Def holding	1,058,548	0	0.003	0.050	0.000	1.000
Off PI	1,058,548	0	0.001	0.034	0.000	1.000
Def PI	1,058,548	0	0.008	0.088	0.000	1.000
Attendance (endogenous)						
Attendance (10k)	1,058,548	0	4.050	2.748	0.000	11.309
Attendance (% obs)	1,058,548	0	0.801	0.248	0.000	1.000
Situation controls						
Home offense	1,058,548	0	0.503	0.500	0.000	1.000
Down (FE when pooled)	1,058,548	0	1.758	0.779	1.000	3.000
Distance	1,058,548	0	8.639	3.850	-8.000	99.000
Yards to goal	1,058,548	0	51.512	24.160	-2.000	100.000
Period	1,058,548	0	2.494	1.117	1.000	11.000
Half sec left	1,058,548	0	910.873	544.284	0.000	3,960.000
Score diff (off)	1,058,548	0	-0.945	14.599	-98.000	98.000
Strength controls						
Off Elo	1,058,548	0	1,506.728	243.765	675.000	2,384.000
Def Elo	1,058,548	0	1,500.581	243.215	675.000	2,384.000
Spread (off)	1,058,548	0	-0.326	14.541	-55.500	55.500
Spread miss	1,058,548	0	0.001	0.031	0.000	1.000
Game controls						
Away dist	1,058,548	0	582.450	529.366	0.000	5,076.458
Away tz chg	1,058,548	0	0.012	0.860	-5.000	5.000
Temp (F)	1,058,548	0	65.198	14.911	8.000	107.900
Wind (mph)	1,058,548	0	7.104	4.070	0.000	27.500
Precip (mm)	1,058,548	0	0.101	0.526	0.000	14.400
Snow (cm)	1,058,548	0	0.001	0.027	0.000	0.980
Conference game	1,058,548	0	0.737	0.441	0.000	1.000
Top vs top	1,058,548	0	0.440	0.496	0.000	1.000

Table 1: Summary statistics: penalty-level model variables (downs 1–3) (*continued*)

Variable	N	NA	Mean	SD	Min	Max
Top vs bottom	1,058,548	0	0.106	0.308	0.000	1.000
Bottom vs top	1,058,548	0	0.030	0.172	0.000	1.000
Team-opp familiarity	1,058,548	0	1.201	0.879	0.000	4.000
Home stadium fam	1,058,548	0	12.506	2.927	0.000	22.000
Away stadium fam	1,058,548	0	0.527	0.546	0.000	13.000
Coach H2H wins	1,058,548	0	0.488	0.904	0.000	9.000
Coach H2H losses	1,058,548	0	0.507	0.920	0.000	10.000
Off win lag 1	1,058,548	0	0.494	0.500	0.000	1.000
Off win lag 2	1,058,548	0	0.495	0.500	0.000	1.000
Off win lag 3	1,058,548	0	0.493	0.500	0.000	1.000
Def win lag 1	1,058,548	0	0.486	0.500	0.000	1.000
Def win lag 2	1,058,548	0	0.490	0.500	0.000	1.000
Def win lag 3	1,058,548	0	0.488	0.500	0.000	1.000
Off pen lag 1	1,058,548	0	4.166	2.565	0.000	19.000
Off pen lag 2	1,058,548	0	4.158	2.575	0.000	19.000
Off pen lag 3	1,058,548	0	4.113	2.602	0.000	19.000
Def pen lag 1	1,058,548	0	4.179	2.568	0.000	19.000
Def pen lag 2	1,058,548	0	4.156	2.582	0.000	19.000
Def pen lag 3	1,058,548	0	4.109	2.602	0.000	19.000
Off form miss 1	1,058,548	0	0.001	0.026	0.000	1.000
Off form miss 2	1,058,548	0	0.001	0.034	0.000	1.000
Off form miss 3	1,058,548	0	0.002	0.043	0.000	1.000
Def form miss 1	1,058,548	0	0.001	0.026	0.000	1.000
Def form miss 2	1,058,548	0	0.001	0.034	0.000	1.000
Def form miss 3	1,058,548	0	0.002	0.043	0.000	1.000
Fixed effects						
Venues (count)	151					
Seasons (count)	11					
Weeks (count)	15					
Days of week (count)	7					
Instrument						
COVID (2020)	1,058,548	0	0.064	0.244	0.000	1.000

Notice that we group penalties into eight outcome variables. Similar to Farnell (2023) we separate nondiscretionary pre-snap penalties, which involve minimal to no referee judgment. These four are:

- (1) False start
- (2) Presnap off (ex FS): all other offensive pre-snap penalties such as delay of game, illegal formation, illegal shift, illegal motion etc
- (3) Defensive offside
- (4) Presnap def (ex offside): all other defensive pre-snap penalties such as encroachment, neutral zone infraction, 12 men on the field, etc.

The next four require some form of judgment from the referee and are thus discretionary:

- (5) Offensive holding
- (6) Defensive holding
- (7) Offensive pass interference
- (8) Defensive pass interference

The penalty data came from the “penalty_detail” and “penalty_play_text” fields in the cfbfastR play-by-play data. For penalties that are ambiguous in terms of which team committed the penalty, such as “holding” or “PI” we match team names and the words offense/defense in the text to determine which team committed the penalty.

The last outcome variable is expected points added (EPA). This is directly provided to us by cfbfastR. EPA measures the change in the offense's expected points from before to after a play, where expected points are estimated via a multinomial logistic regression on down, distance, field position, and time remaining (cfbfastR, 2021), following the framework of Yurko, Ventura, and Horowitz (2019).

4 Methodology

4.1 Game-Level

$$Y_g = \beta A_g + \mathbf{X}_g' \boldsymbol{\gamma} + \delta_s + \Phi_g + \varepsilon_g$$

Here g indexes games. Y_g is the home win indicator or the home point margin. A_g is attendance, in units of 10,000 fans or as a percentage of the stadium's max observed attendance that season. $\mathbf{X}_g$ holds the game controls. δ_s holds the season fixed effects. Φ_g holds the venue, week, and day-of-week fixed effects.

The parameter of interest is β .

Given Cook (2026), we are cautious of endogeneity. In Cook's paper he finds little evidence of this bias existing. However there is a possibility that this is due to a lack of heterogeneity in attendance given by Figure 1. Thus we say that OLS is inconsistent if:

$$\text{Cov}(A_g, \varepsilon_g \mid \mathbf{X}_g, \delta_s, \Phi_g) \neq 0.$$

Cook (2026) treats attendance as simultaneously determined with their outcomes, on the grounds that performance and attendance demand move together. Under that characterization the system is:

$$\begin{aligned} Y_g &= \beta A_g + \mathbf{X}_g' \boldsymbol{\gamma} + \delta_s + \Phi_g + \varepsilon_g \\ A_g &= \lambda Y_g + \mathbf{X}_g' \boldsymbol{\delta} + \delta_s + \Phi_g + \nu_g \end{aligned}$$

Because tickets are sold in advance, the realized outcome of game g cannot enter the attendance decision for game g . What attendance responds to is the pre-game expectation of quality, formed from publicly perceived team strength. This gives the following system:

$$\begin{aligned} Y_g &= \beta A_g + \mathbf{X}_g' \boldsymbol{\gamma} + \theta Q_g + \delta_s + \Phi_g + e_g \\ A_g &= \phi E_g + \mathbf{X}_g' \boldsymbol{\psi} + \delta_s + \Phi_g + w_g \\ E_g &= \rho \tilde{Q}_g + \mathbf{X}_g' \boldsymbol{\omega} + \xi_g \\ \tilde{Q}_g &= Q_g + \eta_g \end{aligned}$$

where Q_g is home team quality and $\tilde{Q}_g$ is perceived quality of a team. E_g is the pre-game expectation of game quality formed from perceived quality. θ is the effect of quality on the outcome, ϕ is the effect of expectations on attendance, and ρ is the effect of perceived quality on expectations. Crucially Y_g does not enter the attendance equation and E_g does not enter the outcome equation. Attendance and the outcome are linked only through the common root Q_g .

Substituting $\tilde{Q}_g$ into E_g , and E_g into the attendance equation, attendance collapses to

$$A_g = \phi \rho Q_g + \mathbf{X}_g' (\boldsymbol{\psi} + \phi \boldsymbol{\omega}) + \delta_s + \Phi_g + (w_g + \phi \rho \eta_g + \phi \xi_g).$$

We know that for our OLS estimator to be consistent we need:

$$\text{Cov}(A_g, e_g \mid \mathbf{X}_g, \delta_s, \Phi_g) = 0$$

We can then substitute in our collapsed attendance equation into the covariance:

$$\text{Cov}(A_g, e_g \mid \dots) = \text{Cov}(\phi\rho Q_g + \mathbf{X}_g'(\psi + \phi\omega) + \delta_s + \Phi_g + (w_g + \phi\rho\eta_g + \phi\xi_g), e_g \mid \dots)$$

We can simplify to:

$$\text{Cov}(A_g, e_g \mid \dots) = \phi\rho \text{Cov}(Q_g, e_g \mid \dots) = 0$$

However, we cannot observe a perfect Q_g in reality. Thus we must rely on our controls in $\mathbf{X}_g$ to capture the effect of Q_g . We will have:

$$\text{Cov}(A_g, \theta Q_g + e_g \mid \mathbf{X}_g, \delta_s, \Phi_g) = \theta\phi\rho \text{Var}(Q_g \mid \mathbf{X}_g, \delta_s, \Phi_g)$$

We know that $\theta\phi\rho \neq 0$. Thus we are consistent only if true quality is fully captured by our controls:

$$\text{Var}(Q_g \mid \mathbf{X}_g, \delta_s, \Phi_g) = 0.$$

This condition means that the true quality Q_g is perfectly predicted by our covariates and fixed effects. When it holds Q_g has no leftover variation as it is fully explained by our controls. In our estimations we do not include Q_g as a regressor as we cannot observe it. We instead rely on the pregame Elo ratings of each team and the venue, season, week, and day-of-week FEs to capture Q_g .

The pregame betting spread is another possible strength control that one could reach for. However we do not use it in our main specifications as the spread may price in the home crowd advantage itself (Cook, 2026). Controlling for it would then absorb part of the very effect we are estimating. Thus we use spread as a robustness check and note that any effect that survives the inclusion of spread is the effect that betting markets do not price in.

To proxy for fan expectations we control for the expectation of the game from a fans perspective such as away team travel, conference games, and top vs top match ups.

A related concern is effects that are related to long standing rivalries between teams. These could raise both attendance and penalty levels for any crowd size which would create a non-causal correlation between the two. We note that we control for this with the Elo ratings of both the home and away teams and the team familiarity counts. On top of this the conference game indicator and coach head-to-head records also capture rivalry match ups as rivals typically share a conference and their coaches face each other repeatedly. The away team distance control would also assist us here to identify teams that are geographically close to each other.

We will also estimate a two-stage least squares (2SLS) model as a robustness check to see if there is any evidence of our OLS estimator being inconsistent. In this we present the instrument of COVID as a shock to attendance similar to Cook (2026). For this to be valid the season FEs are removed.

We also note that under a simultaneity characterization the lagged home attendance would be an available instrument (Reed, 2015). However, we have shown that any threat of endogeneity would come from omitted variables. In this case a lagged attendance instrument is invalid as the lagged value for attendance is contaminated with the same potential omitted variable bias.

We instrument attendance with a vector of instruments $\mathbf{Z}_g$. The first stage regresses attendance on the instruments and all exogenous terms:

$$A_g = \mathbf{Z}_g' \boldsymbol{\pi} + \mathbf{X}_g' \boldsymbol{\rho} + \delta_s + \Phi_g + u_g$$

The second stage replaces attendance with its first-stage fitted value $\hat{A}_g$:

$$Y_g = \beta \hat{A}_g + \mathbf{X}'_g \gamma + \delta_s + \Phi_g + \varepsilon_g$$

Identification requires the instruments to be relevant ($\pi \neq 0$) and to satisfy the exclusion restriction, affecting Y_g only through A_g :

$$\text{Cov}(\mathbf{Z}_g, \varepsilon_g \mid \mathbf{X}_g, \delta_s, \Phi_g) = 0.$$

The instrument set $\mathbf{Z}_g$ contains a single instrument which is a binary indicator for the 2020 COVID season:

$$\mathbf{Z}_g \in \{ \text{COVID} \}$$

The COVID specification drops δ_s , since COVID varies only across seasons. OLS obviously retains it. Standard errors are HC1 at the game level.

4.2 Play-level model

Here p indexes plays on downs one through three. Y_p is a first down indicator, yards gained, EPA, or a penalty type indicator. The play is measured from the offensive perspective. $\mathbf{S}_p$ holds the play situation controls. Thus to get the home and away team effects we interact attendance with the `is_home` binary indicator H_p .

The OLS equation is:

$$Y_p = \beta_1 A_p + \beta_2 (A_p \cdot H_p) + \mathbf{S}'_p \theta + \mathbf{X}'_p \gamma + \delta_s + \Phi_p + \varepsilon_p$$

where the base H_p enters $\mathbf{S}_p$ as a control. Note that all team-level controls switch from home/away team perspective to offense/defense perspective. The only exception being away team travel.

In the case that attendance is endogenous, so both A_p and the interaction $A_p \cdot H_p$ are endogenous. H_p itself is exogenous. With two endogenous regressors we use two sets of instruments. These are base instrument Z_p and its interaction with the `is_home` indicator,

$$\mathbf{Z}_p = \{ Z_p, Z_p \cdot H_p \}$$

Which give the two first stages:

$$A_p = \mathbf{Z}'_p \pi_1 + \mathbf{S}'_p \theta_1 + \mathbf{X}'_p \rho_1 + \delta_s + \Phi_p + u_{1p}$$

$$A_p \cdot H_p = \mathbf{Z}'_p \pi_2 + \mathbf{S}'_p \theta_2 + \mathbf{X}'_p \rho_2 + \delta_s + \Phi_p + u_{2p}$$

The instrument sets Z_p are the same as the game level (COVID), each carried with its H_p interaction. Standard errors are clustered by game. We estimate the model pooled across downs (with down fixed effects) and separately for each of downs one, two, and three.

The first down and yards outcomes are estimated on the sample plays that do not include penalties. The penalty outcomes are estimated on all plays.

The second stage replaces both endogenous regressors with their fitted values $\hat{A}_p$ and $\widehat{A_p \cdot H_p}$:

$$Y_p = \beta_1 \hat{A}_p + \beta_2 \widehat{A_p \cdot H_p} + \mathbf{S}'_p \theta + \mathbf{X}'_p \gamma + \delta_s + \Phi_p + \varepsilon_p$$

4.3 Early Timeout Case

The purpose of this early timeout case of the play level model is to capture the times when a team, due to a miscommunication from crowd noise, avoids a delay of game penalty with a timeout. Calling an unnecessary timeout hurts the team in question as it limits options for later in the game. Since teams almost always use all of their timeouts before the game ends we cannot simply estimate if a team called a timeout on a given play. Rather we must look at if the team called an early timeout that is outside of the final minutes of a game where teams use timeouts to manage the clock. Thus we set our sample to only be the first quarter of each half. This lets us use a natural cut off point that removes all time constraints on a team's decision to call a timeout. A timeout taken in these first quarters of a half reflects pressure on the play clock rather than the game clock.

The outcome is defined over play situations p in periods one and three, downs 1–3:

$$Y_p^{TO} \in \{ \text{offensive timeout, defensive timeout} \}$$

The estimating equation adds only the timeout controls $\mathbf{T}_p$:

$$Y_p^{TO} = \beta_1 A_p + \beta_2 (A_p \cdot H_p) + \mathbf{S}_p' \boldsymbol{\theta} + \mathbf{X}_p' \boldsymbol{\gamma} + \mathbf{T}_p' \boldsymbol{\lambda} + \delta_s + \Phi_p + \varepsilon_p$$

where $\mathbf{T}_p$ contains the early timeout controls. They are simply the counts of each team's early timeouts in each of the last three games. Note that the play type indicator is excluded from $\mathbf{S}_p$ as it is not observed.

Note that the threat of OVB is weaker. As in this case the outcome does not directly indicate the quality of a team but rather if a team has a play calling miscommunication. To absorb any possible effect of coaching quality we include the early timeout counts in $\mathbf{T}_p$, other coach related variables in $\mathbf{X}_p$, and team strength controls.

4.4 Diagnostics

4.4.1 Relevance

At the play level each IV specification has two endogenous regressors, A_p and $A_p \cdot H_p$. A single joint first-stage F statistic can be problematic as strong identification of one endogenous regressor can mask weak identification of the other. We therefore report the conditional first stage F statistic of Sanderson and Windmeijer (2016). This residualizes each endogenous regressor with respect to the other before testing instrument strength. Thus we will have a conditional first stage F statistic for A_p and a conditional first stage F statistic for $A_p \cdot H_p$. At the game level there is a single endogenous regressor. So, the statistic collapses to an ordinary robust first stage F.

We benchmark against the Stock and Yogo (2005) critical values for weak instruments. Our conditional F statistics are substantially larger than any relevant thresholds. Thus weak instruments are not a concern here. This is expected since the 2020 COVID season brought attendance from normal levels of capacity to near zero.

Note that every IV specification is exactly identified as we have one instrument per endogenous regressor. Thus no overidentification test is available to us.

4.4.2 Hausman-Wu test

We test the exogeneity of attendance using the Durbin-Wu-Hausman test (Hausman, 1978). The test asks whether our OLS and IV estimates of attendance differ. It works by adding the first stage residuals to our outcome equation and testing if their coefficients are jointly zero. At the play level both endogenous regressors contribute a residual. Rejection would mean our controls do not fully capture team quality and

that leftover variation in Q_g is still correlated with attendance. Note that our COVID specification drops the season fixed effects so the OLS estimates in this comparison drop them as well.

5 Results

Before reporting coefficients we set up the conversion from per play effects to per game effects. Our play counts are:

Table 2: Average play counts (downs 1–3)

Sample	Plays / offense / game	Plays / game	Games
Penalty-free (PBP)	65.9	131.7	7435
All plays (PBP Pen)	71.2	142.4	7435

Mean attendance is 4.053. A per game effect is the coefficient multiplied by 4.053 and by the plays per offense per game above.

5.1 Game-by-Game Results

See Appendix A for our game-by-game results. We see a lack of stable significance across both penalty and game outcome models. In the cases where a variable is significant for a given specification it is not robust across the others. These unstable results suggest asymmetry when the home team is on offense vs when the away team is on offense. Our following play-level results will address this very issue.

5.2 Play-by-Play Results

5.2.1 Pooled

The below results support ours and Cook’s (2026) arguments about how ELO is the proper strength control to be used. Note that the coefficient for the effect of attendance x home is 0.049 (0.007). This does not change when we remove season FEs and when we use COVID as an instrument we fail to reject our Hausman test as the coefficient only moves to 0.047(0.022). Thus we can say that we fail to find evidence of endogeneity here. On top of that our COVID instrument has conditional F scores of 4768.2 and 6241.1. This allows the IV estimate to still be 0.047(0.022) even without a strength control.

Table 3: Attendance and offensive success (play level). nC excludes 2020, nF drops season FEs, COVID uses the 2020 IV.

	No control				Elo control			
	OLS	OLS nC	OLS nF	COVID	OLS	OLS nC	OLS nF	COVID
Outcome: First down								
<i>Attendance (10k)</i>								
Attendance	0.001*	0.002**	0.000	0.000	0.001	0.001**	0.000	0.000
	(0.000)	(0.001)	(0.000)	(0.001)	(0.000)	(0.001)	(0.000)	(0.001)
Attendance × Home	-0.000**	-0.000*	-0.000**	-0.001	-0.000	-0.000	-0.000	-0.001
	(0.000)	(0.000)	(0.000)	(0.001)	(0.000)	(0.000)	(0.000)	(0.001)
First-stage F				7700.7				6241.1
First-stage F (×Home)				5901.4				4768.2
Hausman (p)				0.649				0.616
N	979,365	917,171	979,365	979,365	979,365	917,171	979,365	979,365
Outcome: Yards gained								

Table 3: Attendance and offensive success (play level). nC excludes 2020, nF drops season FEs, COVID uses the 2020 IV. (*continued*)

	OLS	OLS nC	OLS nF	COVID	OLS	OLS nC	OLS nF	COVID
<i>Attendance (10k)</i>								
Attendance	-0.051*** (0.016)	-0.026 (0.024)	-0.055*** (0.011)	-0.027 (0.018)	-0.032** (0.016)	-0.003 (0.025)	-0.035*** (0.011)	-0.027 (0.018)
Attendance $\times$ Home	0.090*** (0.007)	0.095*** (0.007)	0.090*** (0.007)	0.047** (0.022)	0.049*** (0.007)	0.048*** (0.007)	0.049*** (0.007)	0.047** (0.022)
First-stage F				7700.7				6241.1
First-stage F ($\times$ Home)				5901.4				4768.2
Hausman (p)				0.086				0.665
N	979,365	917,171	979,365	979,365	979,365	917,171	979,365	979,365
Outcome: EPA								
<i>Attendance (10k)</i>								
Attendance	-0.007*** (0.002)	-0.001 (0.003)	-0.007*** (0.002)	-0.002 (0.002)	-0.005** (0.002)	0.000 (0.003)	-0.005*** (0.002)	-0.003 (0.002)
Attendance $\times$ Home	0.011*** (0.001)	0.012*** (0.001)	0.011*** (0.001)	0.003 (0.003)	0.006*** (0.001)	0.006*** (0.001)	0.006*** (0.001)	0.003 (0.003)
First-stage F				7691.5				6229.1
First-stage F ($\times$ Home)				5897.2				4762.6
Hausman (p)				0.024				0.384
N	976,471	914,450	976,471	976,471	976,471	914,450	976,471	976,471
Specification								
Strength control	None	None	None	None	Elo	Elo	Elo	Elo
Venue FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Season FE	Yes	Yes	No	No	Yes	Yes	No	No
Week FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Game controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Situation controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Instrument	—	—	—	COVID	—	—	—	COVID

SE in parentheses. *** $p < .01$, ** $p < .05$, * $p < .10$.

Remember that PBP estimations do not include penalties. Reading the two coefficients together, the away offense loses 0.032 yards and 0.005 EPA per play for each additional 10,000 fans. Note that the sum of the base and interaction terms is 0.017 yards and 0.001 EPA per play, neither of which is significant.

Table 4: Offensive effects per play (per 10k)

Outcome	Away offense	Home offense
Yards gained	-0.0319** (0.0160)	0.0171 (0.0160)
EPA	-0.0050** (0.0023)	0.0007 (0.0022)

SE in parentheses. *** $p < .01$, ** $p < .05$, * $p < .10$.

The 0.049 interaction thus represents the size of the home/away gap and not a gain to the home offense. The crowd effect operates by degrading the visiting offense rather than by lifting the home offense. This is what a noise channel would predict. The home crowd is loudest when the visiting offense is calling plays and quietest when its own offense has the ball.

Applying our mean attendance and play counts:

Table 5: Offensive effects at mean attendance

Outcome	Away offense per game	Home offense per game
Yards gained	-8.513** (4.283)	4.572 (4.275)
EPA	-1.326** (0.601)	0.187 (0.597)

SE in parentheses. *** $p < .01$, ** $p < .05$, * $p < .10$.

Appendix B reports robustness checks using attendance (% obs) as the attendance measure and spread as the strength control.

5.2.2 Down-by-Down Results

Given our results supporting using ELO as a strength control and OLS as the estimation method we can then report the relevant results when we split our data by down.

Table 6: Offensive success by down

	Down 1	Down 2	Down 3
Outcome: First down			
Attendance	-0.0002 (0.0008)	0.0011** (0.0005)	0.0006 (0.0007)
Attendance $\times$ Home	0.0005 (0.0005)	-0.0005* (0.0003)	-0.0011*** (0.0004)
N	444,707	330,018	204,640
Outcome: Yards gained			
Attendance	-0.0314 (0.0222)	-0.0159 (0.0241)	-0.0561* (0.0322)
Attendance $\times$ Home	0.0576*** (0.0108)	0.0447*** (0.0120)	0.0340** (0.0160)
N	444,707	330,018	204,640
Outcome: EPA			
Attendance	-0.0040 (0.0025)	-0.0024 (0.0034)	-0.0110* (0.0062)
Attendance $\times$ Home	0.0065*** (0.0012)	0.0051*** (0.0016)	0.0047 (0.0029)
N	443,407	329,039	204,025
Specification			
Strength control	Elo	Elo	Elo
Venue FE	Yes	Yes	Yes
Season FE	Yes	Yes	Yes
Week FE	Yes	Yes	Yes
Game controls	Yes	Yes	Yes
Situation controls	Yes	Yes	Yes
Instrument	—	—	—

Attendance (10k) SE in parentheses. *** p<.01, ** p<.05, * p<.10.

Obviously we see positive effects for the home teams offense with both EPA and yards gained across all downs with the exception of EPA on 3rd down. What is most interesting though is how the signs flip when we look at if a home team obtained a first down. In that case on 2nd and 3rd down the respective coefficients are -0.0005*(0.0003) and -0.0011(0.0004). This can be explained by (1) a change to a more aggressive strategy by the home team offense or (2) a change to a more conservative strategy by the away team defense. If an offense is more aggressive it is more than likely that they gain more yards on average but the variance of the outcome will increase. This polarization of outcomes would then reduce the number of marginal first down gains.

5.3 Play-By-Play Penalty Results

5.3.1 Pooled

Our wider attendance variation lets us identify effects at the play level that Farnell (2023) could not. We report the spread specifications here as a robustness check alongside Elo. Before going coefficient by coefficient

we see a similarity in the effects when we compare our ELO strength specifications to our spread strength specifications. This points to betting markets not pricing in penalties caused by a larger crowd.

Table 7: Penalties (play level, per 10k). nC excludes 2020, nF drops season FEs, COVID uses the 2020 IV.

	Spread control				Elo control			
	OLS	OLS nC	OLS nF	COVID	OLS	OLS nC	OLS nF	COVID
Outcome: False start								
Attendance	0.0007*** (0.0002)	0.0010*** (0.0003)	0.0005*** (0.0001)	0.0004* (0.0002)	0.0007*** (0.0002)	0.0009*** (0.0003)	0.0004*** (0.0001)	0.0004* (0.0002)
Attendance × Home	-0.0003*** (0.0001)	-0.0003*** (0.0001)	-0.0003*** (0.0001)	-0.0006* (0.0003)	-0.0004*** (0.0001)	-0.0003*** (0.0001)	-0.0004*** (0.0001)	-0.0006** (0.0003)
First-stage F				5360.5				6298.9
First-stage F (×Home)				4054.8				4780.6
Hausman (p)				0.097				0.145
N	1,058,548	991,166	1,058,548	1,058,548	1,058,548	991,166	1,058,548	1,058,548
Outcome: Presnap off (ex FS)								
Attendance	0.0001 (0.0001)	0.0000 (0.0001)	0.0001 (0.0001)	0.0001 (0.0001)	0.0001 (0.0001)	0.0001 (0.0001)	0.0001 (0.0001)	0.0001 (0.0001)
Attendance × Home	-0.0002*** (0.0000)	-0.0002*** (0.0001)	-0.0002*** (0.0000)	-0.0001 (0.0001)	-0.0002*** (0.0000)	-0.0002*** (0.0001)	-0.0002*** (0.0000)	-0.0002 (0.0001)
First-stage F				5360.5				6298.9
First-stage F (×Home)				4054.8				4780.6
Hausman (p)				0.987				0.978
N	1,058,548	991,166	1,058,548	1,058,548	1,058,548	991,166	1,058,548	1,058,548
Outcome: Def offside								
Attendance	-0.0002** (0.0001)	-0.0001 (0.0002)	-0.0003*** (0.0001)	-0.0005*** (0.0001)	-0.0002* (0.0001)	-0.0000 (0.0002)	-0.0003*** (0.0001)	-0.0004*** (0.0001)
Attendance × Home	0.0000 (0.0001)	-0.0000 (0.0001)	0.0000 (0.0001)	0.0003 (0.0002)	-0.0000 (0.0001)	-0.0000 (0.0001)	-0.0000 (0.0001)	0.0003 (0.0002)
First-stage F				5360.5				6298.9
First-stage F (×Home)				4054.8				4780.6
Hausman (p)				0.106				0.114
N	1,058,548	991,166	1,058,548	1,058,548	1,058,548	991,166	1,058,548	1,058,548
Outcome: Presnap def (ex off)								
Attendance	-0.0000 (0.0001)	-0.0001 (0.0001)	-0.0000 (0.0000)	-0.0000 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0000 (0.0000)	-0.0000 (0.0001)
Attendance × Home	0.0000 (0.0000)	-0.0000 (0.0000)	0.0000 (0.0000)	0.0001 (0.0001)	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.0001 (0.0001)
First-stage F				5360.5				6298.9
First-stage F (×Home)				4054.8				4780.6
Hausman (p)				0.374				0.373
N	1,058,548	991,166	1,058,548	1,058,548	1,058,548	991,166	1,058,548	1,058,548
Outcome: Off holding								
Attendance	-0.0002 (0.0001)	-0.0000 (0.0002)	-0.0001 (0.0001)	-0.0001 (0.0002)	-0.0002 (0.0001)	-0.0000 (0.0002)	-0.0001 (0.0001)	-0.0001 (0.0002)
Attendance × Home	-0.0000 (0.0001)	-0.0000 (0.0001)	-0.0000 (0.0001)	0.0002 (0.0002)	0.0000 (0.0001)	0.0000 (0.0001)	0.0000 (0.0001)	0.0002 (0.0002)
First-stage F				5360.5				6298.9
First-stage F (×Home)				4054.8				4780.6
Hausman (p)				0.421				0.490
N	1,058,548	991,166	1,058,548	1,058,548	1,058,548	991,166	1,058,548	1,058,548
Outcome: Def holding								
Attendance	-0.0000 (0.0001)	0.0000 (0.0001)	-0.0001* (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0000 (0.0001)	-0.0001* (0.0001)	-0.0001 (0.0001)
Attendance × Home	-0.0000 (0.0000)	-0.0000 (0.0000)	-0.0000 (0.0000)	-0.0000 (0.0001)	-0.0000 (0.0000)	-0.0000 (0.0000)	-0.0000 (0.0000)	-0.0000 (0.0001)
First-stage F				5360.5				6298.9
First-stage F (×Home)				4054.8				4780.6
Hausman (p)				0.935				0.946
N	1,058,548	991,166	1,058,548	1,058,548	1,058,548	991,166	1,058,548	1,058,548
Outcome: Off PI								
Attendance	0.0000 (0.0000)	-0.0000 (0.0001)	0.0000 (0.0000)	-0.0000 (0.0001)	-0.0000 (0.0000)	-0.0000 (0.0001)	0.0000 (0.0000)	-0.0000 (0.0001)
Attendance × Home	-0.0000 (0.0000)	-0.0000 (0.0000)	-0.0000 (0.0000)	0.0001 (0.0001)	-0.0000 (0.0000)	-0.0000 (0.0000)	-0.0000 (0.0000)	0.0001 (0.0001)
First-stage F				5360.5				6298.9
First-stage F (×Home)				4054.8				4780.6
Hausman (p)				0.677				0.705
N	1,058,548	991,166	1,058,548	1,058,548	1,058,548	991,166	1,058,548	1,058,548
Outcome: Def PI								
Attendance	0.0001 (0.0001)	0.0000 (0.0002)	-0.0002* (0.0001)	-0.0004** (0.0001)	0.0000 (0.0001)	-0.0000 (0.0002)	-0.0002** (0.0001)	-0.0004** (0.0001)
Attendance × Home	0.0001** (0.0001)	0.0001* (0.0001)	0.0001* (0.0001)	0.0002 (0.0002)	0.0002** (0.0001)	0.0001** (0.0001)	0.0002** (0.0001)	0.0002 (0.0002)

Table 7: Penalties (play level, per 10k). nC excludes 2020, nF drops season FEs, COVID uses the 2020 IV. (continued)

	OLS	OLS nC	OLS nF	COVID	OLS	OLS nC	OLS nF	COVID
First-stage F				5360.5				6298.9
First-stage F ($\times$ Home)				4054.8				4780.6
Hausman (p)				0.041				0.049
N	1,058,548	991,166	1,058,548	1,058,548	1,058,548	991,166	1,058,548	1,058,548
Specification								
Strength control	Spread	Spread	Spread	Spread	Elo	Elo	Elo	Elo
Venue FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Season FE	Yes	Yes	No	No	Yes	Yes	No	No
Week FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Game controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Situation controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Instrument	—	—	—	COVID	—	—	—	COVID

Attendance (10k). SE in parentheses. *** $p < .01$, ** $p < .05$, * $p < .10$.

We see that away team offenses commit 0.0007 (0.0002) more false starts for each 10,000 fans in attendance. However the coefficient of the interaction term with home is -0.0004(0.0001). The sum of these two indicates that home teams do false start more as the crowd grows. However it is at a lower rate than the away team. Similarly, offensive teams when at home have -0.0002(0.0000) less “other pre-snap” penalties per-play relative to away team offenses.

We find that home team defenses jump offside less when the crowds are larger as the effect of the base attendance variable on defensive offsides is -0.0002(0.0001) per 10,000 fans. This aligns with Farnell’s (2023) argument that fans are louder when the opposing team has the ball. Thus it forces the defense to not attempt to anticipate the snap. When they don’t rely on what they hear they can’t be tricked by a quarterback’s snap count as often.

Applying our mean attendance and play counts:

Table 8: Penalty effects at mean attendance

Outcome	Away offense per game	Home offense per game
False start	0.194*** (0.051)	0.089* (0.049)
Def offside	-0.061* (0.033)	-0.063* (0.034)

SE in parentheses. *** $p < .01$, ** $p < .05$, * $p < .10$.

Interestingly we see that when the home team is on offense the probability of defensive pass interference being called (which is committed by the away team) rises 0.0002(0.0001) per 10,000 fans. This could be through the channel of referee bias. A second theory stems from our discussion on how attendance coefficients for first down and yards gained have differing signs. The home team is more aggressive and attempts longer passes. In doing so they gain more yards, increase the variance of their plays which lowers first down completion rates, and increases the probability of drawing a defensive pass interference call.

5.3.2 Down-by-Down

When we split by downs we see expected patterns. On first downs the directions remain the same but magnitudes increase as the down does. For example on third down the away team false starts 0.0014(0.0004) per 10,000 fans in attendance. The effect almost triples which aligns with noise being a channel in the mechanism as crowds are famously louder on third downs due to their importance.

Table 9: Penalties by down (per 10k)

	Down 1	Down 2	Down 3
Outcome: False start			
Attendance	0.0005**	0.0004	0.0014***

Table 9: Penalties by down (per 10k) (*continued*)

	Down 1	Down 2	Down 3
	(0.0002)	(0.0003)	(0.0004)
Attendance $\times$ Home	-0.0001	-0.0004***	-0.0008***
	(0.0001)	(0.0002)	(0.0002)
N	480,476	353,511	224,561
Outcome: Presnap off (ex FS)			
Attendance	0.0001	-0.0000	0.0003
	(0.0001)	(0.0001)	(0.0002)
Attendance $\times$ Home	-0.0002***	-0.0001	-0.0002
	(0.0001)	(0.0001)	(0.0001)
N	480,476	353,511	224,561
Outcome: Def offside			
Attendance	-0.0004**	-0.0000	-0.0002
	(0.0002)	(0.0002)	(0.0003)
Attendance $\times$ Home	-0.0000	-0.0000	0.0000
	(0.0001)	(0.0001)	(0.0002)
N	480,476	353,511	224,561
Outcome: Presnap def (ex off)			
Attendance	0.0000	-0.0001	-0.0001
	(0.0001)	(0.0001)	(0.0001)
Attendance $\times$ Home	0.0000	-0.0000	0.0000
	(0.0000)	(0.0000)	(0.0001)
N	480,476	353,511	224,561
Outcome: Off holding			
Attendance	-0.0003	-0.0002	-0.0002
	(0.0002)	(0.0002)	(0.0002)
Attendance $\times$ Home	0.0001	-0.0001	0.0000
	(0.0001)	(0.0001)	(0.0001)
N	480,476	353,511	224,561
Outcome: Def holding			
Attendance	-0.0001	-0.0000	0.0000
	(0.0001)	(0.0001)	(0.0002)
Attendance $\times$ Home	-0.0000	0.0000	-0.0001
	(0.0000)	(0.0001)	(0.0001)
N	480,476	353,511	224,561
Outcome: Off PI			
Attendance	-0.0000	0.0001	-0.0000
	(0.0001)	(0.0001)	(0.0001)
Attendance $\times$ Home	0.0000	-0.0000	0.0000
	(0.0000)	(0.0000)	(0.0001)
N	480,476	353,511	224,561
Outcome: Def PI			
Attendance	0.0002	-0.0002	0.0001
	(0.0002)	(0.0002)	(0.0003)
Attendance $\times$ Home	0.0001	0.0002*	0.0003
	(0.0001)	(0.0001)	(0.0002)
N	480,476	353,511	224,561
Specification			
Strength control	Elo	Elo	Elo
Venue FE	Yes	Yes	Yes
Season FE	Yes	Yes	Yes
Week FE	Yes	Yes	Yes
Game controls	Yes	Yes	Yes
Situation controls	Yes	Yes	Yes
Instrument	—	—	—

Attendance (10k) SE in parentheses. *** p<.01, ** p<.05, * p<.10.

5.4 Timeouts

Our timeout specifications provide further support for the noise channel. Away offenses call significantly more early timeouts as attendance rises, with the effect roughly doubling on third down, mirroring the pattern found for false starts. Home offenses also call more early timeouts, though at roughly half the away team rate. Defensive timeouts show no attendance effect, consistent with noise pressuring the offense's play clock rather than the defense.

Table 10: Early timeouts (periods 1 & 3, per 10k)

	Pooled	Down 1	Down 2	Down 3
Outcome: Offensive timeout				
Attendance	0.0009*** (0.0002)	0.0005* (0.0003)	0.0008*** (0.0003)	0.0021*** (0.0007)
Attendance $\times$ Home	-0.0004*** (0.0001)	-0.0003 (0.0002)	-0.0006*** (0.0002)	-0.0005 (0.0003)
N	481,290	220,488	161,077	99,725
Outcome: Defensive timeout				
Attendance	-0.0002 (0.0002)	-0.0003 (0.0002)	-0.0000 (0.0002)	-0.0004 (0.0004)
Attendance $\times$ Home	0.0001 (0.0001)	0.0002 (0.0001)	0.0001 (0.0001)	0.0001 (0.0002)
N	481,290	220,488	161,077	99,725
Specification				
Strength control	Elo	Elo	Elo	Elo
Venue FE	Yes	Yes	Yes	Yes
Season FE	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes
Game controls	Yes	Yes	Yes	Yes
Situation controls	Yes	Yes	Yes	Yes
Instrument	—	—	—	—

Attendance (10k) SE in parentheses. *** p<.01, ** p<.05, * p<.10.

An early timeout is a resource that can be spent to avoid a yardage loss. The away offense spends them at a higher rate as the crowd grows. Thus in the later portion of the game or half the away team will not be able to manage the clock as well as the home team.

6 Conclusion

Given our results we flag that as crowds increase the gap between the home and away offense widens outside of channels like referee bias. We reveal that for each 10,000 fans the home offense gains an estimated 0.049 yards and 0.006 EPA per play over the away offense. As shown above this gap comes from the away offense declining and not the home offense improving.

The effect is asymmetric. When the away team has the ball their offense loses an estimated 8.513 yards and 1.326 expected points per game at mean attendance. The home-away gap is 0.049 yards and 0.006 EPA per play. This asymmetry is what a noise channel would predict since home crowds are loudest when the visiting offense has the ball and quietest when their own offense is calling plays.

We find evidence similar to Farnell (2023) that the crowd affects nondiscretionary penalties. The evidence is most concentrated within offensive false starts and defensive offsides.

We answer an unresolved question from Farnell (2023) that is which team false starts more due to the crowd. Our answer is that the away team does so due to the noise channel. This was revealed by the increase in effect magnitude found on 3rd down. With Cook’s paper we have confirmed that ELO is a valid control and that the omitted variable bias is not strong enough to call for the use of an instrumental variable.

7 References

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A Game-Level Results

These estimations have very little significance on a game-by-game level.

Table 11: Attendance and home advantage

	Spread control				Elo control			
	OLS	OLS nC	OLS nF	COVID	OLS	OLS nC	OLS nF	COVID
Outcome: Home win								
<i>Attendance (10k)</i>								
Attendance	0.006 (0.007)	-0.001 (0.010)	-0.000 (0.004)	-0.003 (0.006)	0.006 (0.007)	0.000 (0.010)	0.002 (0.005)	0.001 (0.006)
First-stage F				1371.0				1383.1
Hausman (p)				0.421				0.731
N	7,483	7,002	7,483	7,483	7,483	7,002	7,483	7,483
Outcome: Home margin								
<i>Attendance (10k)</i>								
Attendance	-0.034 (0.260)	-0.490 (0.385)	0.043 (0.168)	0.128 (0.219)	0.135 (0.267)	-0.004 (0.404)	0.254 (0.172)	0.345 (0.226)
First-stage F				1371.0				1383.1
Hausman (p)				0.547				0.527
N	7,483	7,002	7,483	7,483	7,483	7,002	7,483	7,483
Specification								
Strength control	Spread	Spread	Spread	Spread	Elo	Elo	Elo	Elo
Venue FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Season FE	Yes	Yes	No	No	Yes	Yes	No	No
Week FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Game controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Situation controls	No	No	No	No	No	No	No	No
Instrument	—	—	—	COVID	—	—	—	COVID

SE in parentheses. *** p<.01, ** p<.05, * p<.10.

We see alignment of our results with false starts and our main OLS specification. Note however that a lack of season fixed effects drives significance on the home team estimations of defensive offsides.

Table 12: Home-team penalties (game level, per 10k)

	Spread control				Elo control			
	OLS	OLS nC	OLS nF	COVID	OLS	OLS nC	OLS nF	COVID
Outcome: False start								
Attendance	0.0358** (0.0176)	0.0307 (0.0285)	0.0034 (0.0117)	-0.0207 (0.0166)	0.0341* (0.0180)	0.0246 (0.0299)	0.0002 (0.0119)	-0.0228 (0.0166)
First-stage F				1371.0				1383.1
Hausman (p)				0.024				0.029
N	7,483	7,002	7,483	7,483	7,483	7,002	7,483	7,483
Outcome: Presnap off (ex FS)								
Attendance	-0.0132 (0.0109)	-0.0207 (0.0158)	-0.0153** (0.0071)	-0.0174* (0.0092)	-0.0134 (0.0113)	-0.0232 (0.0168)	-0.0160** (0.0072)	-0.0182** (0.0092)
First-stage F				1371.0				1383.1
Hausman (p)				0.713				0.693
N	7,483	7,002	7,483	7,483	7,483	7,002	7,483	7,483
Outcome: Def offside								
Attendance	-0.0150 (0.0120)	0.0003 (0.0169)	-0.0275*** (0.0082)	-0.0348*** (0.0110)	-0.0166 (0.0124)	-0.0036 (0.0178)	-0.0282*** (0.0083)	-0.0344*** (0.0110)
First-stage F				1371.0				1383.1
Hausman (p)				0.263				0.340
N	7,483	7,002	7,483	7,483	7,483	7,002	7,483	7,483
Outcome: Presnap def (ex off)								
Attendance	-0.0003 (0.0054)	-0.0067 (0.0093)	0.0020 (0.0033)	0.0030 (0.0040)	-0.0014 (0.0057)	-0.0106 (0.0103)	0.0014 (0.0034)	0.0028 (0.0040)
First-stage F				1371.0				1383.1
Hausman (p)				0.739				0.648

Table 12: Home-team penalties (game level, per 10k) (*continued*)

	OLS	OLS nC	OLS nF	COVID	OLS	OLS nC	OLS nF	COVID
N	7,483	7,002	7,483	7,483	7,483	7,002	7,483	7,483
Outcome: Off holding								
Attendance	-0.0325** (0.0159)	-0.0254 (0.0240)	-0.0182* (0.0105)	-0.0086 (0.0137)	-0.0305* (0.0162)	-0.0244 (0.0251)	-0.0180* (0.0106)	-0.0098 (0.0137)
First-stage F				1371.0				1383.1
Hausman (p)				0.293				0.362
N	7,483	7,002	7,483	7,483	7,483	7,002	7,483	7,483
Outcome: Def holding								
Attendance	0.0037 (0.0074)	0.0119 (0.0109)	-0.0030 (0.0049)	-0.0062 (0.0058)	0.0017 (0.0075)	0.0080 (0.0112)	-0.0042 (0.0049)	-0.0065 (0.0058)
First-stage F				1371.0				1383.1
Hausman (p)				0.442				0.567
N	7,483	7,002	7,483	7,483	7,483	7,002	7,483	7,483
Outcome: Off PI								
Attendance	0.0016 (0.0046)	-0.0034 (0.0076)	0.0031 (0.0028)	0.0031 (0.0041)	0.0011 (0.0046)	-0.0047 (0.0078)	0.0029 (0.0028)	0.0031 (0.0042)
First-stage F				1371.0				1383.1
Hausman (p)				0.989				0.957
N	7,483	7,002	7,483	7,483	7,483	7,002	7,483	7,483
Outcome: Def PI								
Attendance	0.0141 (0.0122)	0.0092 (0.0191)	-0.0080 (0.0081)	-0.0255** (0.0113)	0.0135 (0.0126)	0.0063 (0.0205)	-0.0091 (0.0082)	-0.0259** (0.0113)
First-stage F				1371.0				1383.1
Hausman (p)				0.011				0.014
N	7,483	7,002	7,483	7,483	7,483	7,002	7,483	7,483
Specification								
Strength control	Spread	Spread	Spread	Spread	Elo	Elo	Elo	Elo
Venue FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Season FE	Yes	Yes	No	No	Yes	Yes	No	No
Week FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Game controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Situation controls	No	No	No	No	No	No	No	No
Instrument	—	—	—	COVID	—	—	—	COVID

Attendance (10k). SE in parentheses. *** p<.01, ** p<.05, * p<.10.

Table 13: Away-team penalties (game level, per 10k)

	Spread control				Elo control			
	OLS	OLS nC	OLS nF	COVID	OLS	OLS nC	OLS nF	COVID
Outcome: False start								
Attendance	0.0356** (0.0178)	0.0806*** (0.0283)	0.0269** (0.0124)	0.0239 (0.0173)	0.0282 (0.0181)	0.0622** (0.0299)	0.0227* (0.0124)	0.0230 (0.0173)
First-stage F				1371.0				1383.1
Hausman (p)				0.795				0.978
N	7,483	7,002	7,483	7,483	7,483	7,002	7,483	7,483
Outcome: Presnap off (ex FS)								
Attendance	0.0085 (0.0115)	0.0193 (0.0170)	0.0034 (0.0076)	-0.0020 (0.0097)	0.0095 (0.0118)	0.0223 (0.0177)	0.0036 (0.0077)	-0.0022 (0.0097)
First-stage F				1371.0				1383.1
Hausman (p)				0.424				0.390
N	7,483	7,002	7,483	7,483	7,483	7,002	7,483	7,483
Outcome: Def offside								
Attendance	-0.0277** (0.0125)	-0.0269 (0.0185)	-0.0196** (0.0083)	-0.0109 (0.0106)	-0.0234* (0.0128)	-0.0195 (0.0194)	-0.0187** (0.0084)	-0.0124 (0.0106)
First-stage F				1371.0				1383.1
Hausman (p)				0.200				0.346
N	7,483	7,002	7,483	7,483	7,483	7,002	7,483	7,483
Outcome: Presnap def (ex off)								
Attendance	-0.0058 (0.0050)	-0.0015 (0.0077)	-0.0007 (0.0034)	0.0030 (0.0041)	-0.0074 (0.0050)	-0.0056 (0.0080)	-0.0015 (0.0034)	0.0028 (0.0041)
First-stage F				1371.0				1383.1
Hausman (p)				0.222				0.154
N	7,483	7,002	7,483	7,483	7,483	7,002	7,483	7,483
Outcome: Off holding								
Attendance	0.0018 (0.0144)	0.0241 (0.0235)	-0.0040 (0.0094)	-0.0080 (0.0126)	0.0051 (0.0147)	0.0306 (0.0247)	-0.0044 (0.0095)	-0.0102 (0.0126)
First-stage F				1371.0				1383.1
Hausman (p)				0.650				0.505
N	7,483	7,002	7,483	7,483	7,483	7,002	7,483	7,483

Table 13: Away-team penalties (game level, per 10k) (*continued*)

	OLS	OLS nC	OLS nF	COVID	OLS	OLS nC	OLS nF	COVID
Outcome: Def holding								
Attendance	-0.0049 (0.0070)	0.0011 (0.0107)	-0.0077 (0.0051)	-0.0083 (0.0067)	-0.0059 (0.0072)	-0.0013 (0.0115)	-0.0085 (0.0052)	-0.0086 (0.0067)
First-stage F				1371.0				1383.1
Hausman (p)				0.891				0.964
N	7,483	7,002	7,483	7,483	7,483	7,002	7,483	7,483
Outcome: Off PI								
Attendance	0.0013 (0.0045)	0.0005 (0.0071)	-0.0005 (0.0031)	-0.0024 (0.0045)	0.0004 (0.0046)	-0.0017 (0.0076)	-0.0010 (0.0032)	-0.0025 (0.0045)
First-stage F				1371.0				1383.1
Hausman (p)				0.467				0.558
N	7,483	7,002	7,483	7,483	7,483	7,002	7,483	7,483
Outcome: Def PI								
Attendance	0.0038 (0.0122)	0.0004 (0.0182)	-0.0108 (0.0082)	-0.0215* (0.0112)	0.0031 (0.0125)	-0.0027 (0.0193)	-0.0124 (0.0083)	-0.0228** (0.0112)
First-stage F				1371.0				1383.1
Hausman (p)				0.145				0.151
N	7,483	7,002	7,483	7,483	7,483	7,002	7,483	7,483
Specification								
Strength control	Spread	Spread	Spread	Spread	Elo	Elo	Elo	Elo
Venue FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Season FE	Yes	Yes	No	No	Yes	Yes	No	No
Week FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Game controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Situation controls	No	No	No	No	No	No	No	No
Instrument	—	—	—	COVID	—	—	—	COVID

Attendance (10k). SE in parentheses. *** p<.01, ** p<.05, * p<.10.

B Robustness Checks

B.1 Summary Statistics

Table 14: Summary statistics: game-level model variables

Variable	N	NA	Mean	SD	Min	Max
Outcomes						
Home win	7,487	0	0.584	0.493	0.000	1.000
Home margin	7,487	0	4.186	21.067	-78.000	74.000
Attendance (endogenous)						
Attendance (10k)	7,487	0	4.049	2.761	0.000	11.309
Attendance (% obs)	7,487	0	0.800	0.249	0.000	1.000
Strength controls						
Home Elo	7,487	0	1,515.577	244.509	675.000	2,343.000
Away Elo	7,487	0	1,489.946	243.082	701.000	2,384.000
Spread	7,487	0	-4.520	13.882	-55.500	52.000
Spread miss	7,487	0	0.008	0.088	0.000	1.000
Game controls						
Away dist	7,487	0	584.229	531.734	0.000	5,076.458
Away tz chg	7,487	0	0.013	0.864	-5.000	5.000
Temp (F)	7,487	0	65.013	15.008	8.000	107.900
Wind (mph)	7,487	0	7.117	4.083	0.000	27.500
Precip (mm)	7,487	0	0.104	0.533	0.000	14.400
Snow (cm)	7,487	0	0.001	0.027	0.000	0.980
Conference game	7,487	0	0.737	0.440	0.000	1.000
Top vs top	7,487	0	0.438	0.496	0.000	1.000
Top vs bottom	7,487	0	0.107	0.310	0.000	1.000
Bottom vs top	7,487	0	0.030	0.170	0.000	1.000
Team-opp familiarity	7,487	0	1.199	0.880	0.000	4.000
Home stadium fam	7,487	0	12.525	2.937	0.000	22.000
Away stadium fam	7,487	0	0.526	0.546	0.000	13.000

Table 14: Summary statistics: game-level model variables (*continued*)

Variable	N	NA	Mean	SD	Min	Max
Coach H2H wins	7,487	0	0.490	0.909	0.000	9.000
Coach H2H losses	7,487	0	0.509	0.925	0.000	10.000
Home win lag 1	7,487	0	0.476	0.499	0.000	1.000
Home win lag 2	7,487	0	0.504	0.500	0.000	1.000
Home win lag 3	7,487	0	0.495	0.500	0.000	1.000
Away win lag 1	7,487	0	0.504	0.500	0.000	1.000
Away win lag 2	7,487	0	0.481	0.500	0.000	1.000
Away win lag 3	7,487	0	0.486	0.500	0.000	1.000
Home pen lag 1	7,487	0	4.212	2.562	0.000	19.000
Home pen lag 2	7,487	0	4.158	2.559	0.000	16.000
Home pen lag 3	7,487	0	4.081	2.617	0.000	19.000
Away pen lag 1	7,487	0	4.106	2.546	0.000	16.000
Away pen lag 2	7,487	0	4.142	2.590	0.000	19.000
Away pen lag 3	7,487	0	4.125	2.579	0.000	16.000
Home form miss 1	7,487	0	0.001	0.023	0.000	1.000
Home form miss 2	7,487	0	0.001	0.028	0.000	1.000
Home form miss 3	7,487	0	0.001	0.035	0.000	1.000
Away form miss 1	7,487	0	0.001	0.028	0.000	1.000
Away form miss 2	7,487	0	0.002	0.040	0.000	1.000
Away form miss 3	7,487	0	0.003	0.050	0.000	1.000
Fixed effects						
Venues (count)	151					
Seasons (count)	11					
Weeks (count)	16					
Days of week (count)	7					
Instrument						
COVID (2020)	7,487	0	0.064	0.245	0.000	1.000
Home penalties						
False start	7,487	0	1.164	1.216	0.000	10.000
Presnap off (ex FS)	7,487	0	0.391	0.654	0.000	4.000
Def offside	7,487	0	0.440	0.733	0.000	5.000
Presnap def (ex offside)	7,487	0	0.103	0.335	0.000	5.000
Off holding	7,487	0	0.932	1.027	0.000	8.000
Def holding	7,487	0	0.193	0.460	0.000	4.000
Off PI	7,487	0	0.087	0.307	0.000	4.000
Def PI	7,487	0	0.549	0.775	0.000	5.000
Away penalties						
False start	7,487	0	1.330	1.312	0.000	10.000
Presnap off (ex FS)	7,487	0	0.507	0.762	0.000	6.000
Def offside	7,487	0	0.452	0.766	0.000	7.000
Presnap def (ex offside)	7,487	0	0.109	0.339	0.000	4.000
Off holding	7,487	0	0.878	0.982	0.000	7.000
Def holding	7,487	0	0.193	0.468	0.000	5.000
Off PI	7,487	0	0.085	0.299	0.000	3.000
Def PI	7,487	0	0.604	0.830	0.000	6.000

Table 15: Summary statistics: play-level model variables (downs 1–3)

Variable	N	NA	Mean	SD	Min	Max
Outcomes						
First down	979,365	0	0.288	0.453	0.000	1.000
Yards gained	979,365	0	5.967	9.735	-98.000	99.000
EPA	976,471	2,894	0.091	1.325	-7.505	11.105
Attendance (endogenous)						
Attendance (10k)	979,365	0	4.053	2.750	0.000	11.309
Attendance (% obs)	979,365	0	0.801	0.248	0.000	1.000
Situation controls						
Home offense	979,365	0	0.503	0.500	0.000	1.000

Table 15: Summary statistics: play-level model variables (downs 1–3) (*continued*)

Variable	N	NA	Mean	SD	Min	Max
Pass play	979,365	0	0.481	0.500	0.000	1.000
Down (FE when pooled)	979,365	0	1.755	0.776	1.000	3.000
Distance	979,365	0	8.624	3.801	-8.000	99.000
Yards to goal	979,365	0	51.582	24.153	0.000	100.000
Period	979,365	0	2.493	1.118	1.000	11.000
Half sec left	979,365	0	911.429	545.040	0.000	3,960.000
Score diff (off)	979,365	0	-0.889	14.634	-98.000	98.000
Strength controls						
Off Elo	979,365	0	1,507.396	243.909	675.000	2,384.000
Def Elo	979,365	0	1,500.438	243.307	675.000	2,384.000
Spread (off)	979,365	0	-0.364	14.552	-55.500	55.500
Spread miss	979,365	0	0.001	0.031	0.000	1.000
Game controls						
Away tz chg	979,365	0	0.013	0.860	-5.000	5.000
Temp (F)	979,365	0	65.166	14.927	8.000	107.900
Wind (mph)	979,365	0	7.109	4.072	0.000	27.500
Precip (mm)	979,365	0	0.101	0.526	0.000	14.400
Snow (cm)	979,365	0	0.001	0.027	0.000	0.980
Conference game	979,365	0	0.737	0.440	0.000	1.000
Top vs top	979,365	0	0.440	0.496	0.000	1.000
Top vs bottom	979,365	0	0.107	0.309	0.000	1.000
Bottom vs top	979,365	0	0.030	0.171	0.000	1.000
Team-opp familiarity	979,365	0	1.201	0.879	0.000	4.000
Home stadium fam	979,365	0	12.512	2.928	0.000	22.000
Away stadium fam	979,365	0	0.527	0.546	0.000	13.000
Coach H2H wins	979,365	0	0.489	0.905	0.000	9.000
Coach H2H losses	979,365	0	0.508	0.921	0.000	10.000
Off win lag 1	979,365	0	0.494	0.500	0.000	1.000
Off win lag 2	979,365	0	0.496	0.500	0.000	1.000
Off win lag 3	979,365	0	0.494	0.500	0.000	1.000
Def win lag 1	979,365	0	0.486	0.500	0.000	1.000
Def win lag 2	979,365	0	0.490	0.500	0.000	1.000
Def win lag 3	979,365	0	0.488	0.500	0.000	1.000
Off pen lag 1	979,365	0	4.161	2.563	0.000	19.000
Off pen lag 2	979,365	0	4.154	2.575	0.000	19.000
Off pen lag 3	979,365	0	4.109	2.600	0.000	19.000
Def pen lag 1	979,365	0	4.175	2.565	0.000	19.000
Def pen lag 2	979,365	0	4.154	2.581	0.000	19.000
Def pen lag 3	979,365	0	4.108	2.601	0.000	19.000
Off form miss 1	979,365	0	0.001	0.026	0.000	1.000
Off form miss 2	979,365	0	0.001	0.034	0.000	1.000
Off form miss 3	979,365	0	0.002	0.043	0.000	1.000
Def form miss 1	979,365	0	0.001	0.026	0.000	1.000
Def form miss 2	979,365	0	0.001	0.034	0.000	1.000
Def form miss 3	979,365	0	0.002	0.043	0.000	1.000
Fixed effects						
Venues (count)	151					
Seasons (count)	11					
Weeks (count)	15					
Days of week (count)	7					
Instrument						
COVID (2020)	979,365	0	0.064	0.244	0.000	1.000

Table 16: Summary statistics: early-timeout model variables (periods 1 & 3, downs 1–3)

Variable	N	NA	Mean	SD	Min	Max
Outcomes						
Offensive timeout	481,290	0	0.010	0.098	0.000	1.000

Table 16: Summary statistics: early-timeout model variables (periods 1 & 3, downs 1–3) (*continued*)

Variable	N	NA	Mean	SD	Min	Max
Defensive timeout	481,290	0	0.005	0.072	0.000	1.000
Attendance (endogenous)						
Attendance (10k)	481,290	0	4.060	2.754	0.000	11.309
Attendance (% obs)	481,290	0	0.802	0.247	0.000	1.000
Situation controls						
Home offense	481,290	0	0.502	0.500	0.000	1.000
Down (FE when pooled)	481,290	0	1.749	0.776	1.000	3.000
Distance	481,290	0	8.598	3.759	0.000	93.000
Yards to goal	481,290	0	52.865	23.795	0.000	103.000
Period	481,290	0	1.988	1.000	1.000	3.000
Half sec left	481,290	0	1,394.463	269.691	900.000	3,960.000
Score diff (off)	481,290	0	-0.762	12.757	-76.000	76.000
Strength controls						
Off Elo	481,290	0	1,507.908	244.387	675.000	2,384.000
Def Elo	481,290	0	1,499.872	243.959	675.000	2,384.000
Spread (off)	481,290	0	-0.424	14.663	-55.500	55.500
Spread miss	481,290	0	0.001	0.032	0.000	1.000
Game controls						
Away dist	481,290	0	581.727	528.302	0.000	5,076.458
Away tz chg	481,290	0	0.013	0.859	-5.000	5.000
Temp (F)	481,290	0	65.193	14.929	8.000	107.900
Wind (mph)	481,290	0	7.105	4.071	0.000	27.500
Precip (mm)	481,290	0	0.102	0.531	0.000	14.400
Snow (cm)	481,290	0	0.001	0.027	0.000	0.980
Conference game	481,290	0	0.735	0.441	0.000	1.000
Top vs top	481,290	0	0.439	0.496	0.000	1.000
Top vs bottom	481,290	0	0.108	0.310	0.000	1.000
Bottom vs top	481,290	0	0.030	0.172	0.000	1.000
Team-opp familiarity	481,290	0	1.200	0.880	0.000	4.000
Home stadium fam	481,290	0	12.509	2.928	0.000	22.000
Away stadium fam	481,290	0	0.526	0.544	0.000	13.000
Coach H2H wins	481,290	0	0.488	0.903	0.000	9.000
Coach H2H losses	481,290	0	0.505	0.918	0.000	10.000
Off win lag 1	481,290	0	0.495	0.500	0.000	1.000
Off win lag 2	481,290	0	0.496	0.500	0.000	1.000
Off win lag 3	481,290	0	0.494	0.500	0.000	1.000
Def win lag 1	481,290	0	0.484	0.500	0.000	1.000
Def win lag 2	481,290	0	0.489	0.500	0.000	1.000
Def win lag 3	481,290	0	0.488	0.500	0.000	1.000
Off pen lag 1	481,290	0	4.159	2.563	0.000	19.000
Off pen lag 2	481,290	0	4.153	2.579	0.000	19.000
Off pen lag 3	481,290	0	4.112	2.604	0.000	19.000
Def pen lag 1	481,290	0	4.177	2.565	0.000	19.000
Def pen lag 2	481,290	0	4.161	2.584	0.000	19.000
Def pen lag 3	481,290	0	4.109	2.601	0.000	19.000
Off form miss 1	481,290	0	0.001	0.027	0.000	1.000
Off form miss 2	481,290	0	0.001	0.035	0.000	1.000
Off form miss 3	481,290	0	0.002	0.043	0.000	1.000
Def form miss 1	481,290	0	0.001	0.025	0.000	1.000
Def form miss 2	481,290	0	0.001	0.034	0.000	1.000
Def form miss 3	481,290	0	0.002	0.043	0.000	1.000
Off TO early lag 1	481,290	0	0.592	0.947	0.000	10.000
Off TO early lag 2	481,290	0	0.585	0.939	0.000	10.000
Off TO early lag 3	481,290	0	0.577	0.926	0.000	10.000
Def TO early lag 1	481,290	0	0.591	0.949	0.000	10.000
Def TO early lag 2	481,290	0	0.585	0.942	0.000	10.000
Def TO early lag 3	481,290	0	0.578	0.927	0.000	10.000
Off TO early miss 1	481,290	0	0.006	0.079	0.000	1.000
Off TO early miss 2	481,290	0	0.013	0.111	0.000	1.000
Off TO early miss 3	481,290	0	0.022	0.146	0.000	1.000
Def TO early miss 1	481,290	0	0.006	0.079	0.000	1.000

Table 16: Summary statistics: early-timeout model variables (periods 1 & 3, downs 1–3) (*continued*)

Variable	N	NA	Mean	SD	Min	Max
Def TO early miss 2	481,290	0	0.013	0.111	0.000	1.000
Def TO early miss 3	481,290	0	0.022	0.147	0.000	1.000
Fixed effects						
Venues (count)	151					
Seasons (count)	11					
Weeks (count)	15					
Days of week (count)	7					
Instrument						
COVID (2020)	481,290	0	0.063	0.243	0.000	1.000

B.2 PBP

Again most significances here match directions of our attendance (10k) coefficients in our main PBP results.

B.2.1 Pooled

Table 17: Attendance and offensive success (play level, % obs)

	No control				Spread control				Elo control			
	OLS	OLS nC	OLS nF	COVID	OLS	OLS nC	OLS nF	COVID	OLS	OLS nC	OLS nF	COVID
Outcome: First down												
Attendance (% obs)												
Attendance	0.010*** (0.003)	0.011*** (0.003)	0.003 (0.003)	0.001 (0.003)	0.009*** (0.003)	0.010*** (0.003)	0.003 (0.003)	0.000 (0.003)	0.009*** (0.003)	0.010*** (0.003)	0.003 (0.003)	0.001 (0.003)
Attendance × Home	-0.003 (0.003)	-0.003 (0.004)	-0.003 (0.003)	-0.004 (0.004)	-0.002 (0.003)	-0.001 (0.004)	-0.002 (0.003)	-0.003 (0.004)	-0.002 (0.003)	-0.002 (0.004)	-0.002 (0.003)	-0.004 (0.004)
First-stage F				15485.2				15269.3				15380.7
First-stage F (×Home)				15212.5				14935.5				15040.8
Hausman (p)				0.028				0.026				0.033
N	979,365	917,171	979,365	979,365	979,365	917,171	979,365	979,365	979,365	917,171	979,365	979,365
Outcome: Yards gained												
Attendance (% obs)												
Attendance	-0.052 (0.106)	0.034 (0.119)	-0.094 (0.071)	-0.135 (0.094)	0.040 (0.106)	0.165 (0.118)	-0.001 (0.070)	-0.076 (0.092)	-0.018 (0.107)	0.124 (0.120)	-0.053 (0.071)	-0.137 (0.095)
Attendance × Home	0.167** (0.077)	0.078 (0.111)	0.167** (0.077)	0.239** (0.109)	-0.016 (0.075)	-0.185* (0.109)	-0.016 (0.075)	0.125 (0.108)	0.078 (0.076)	-0.118 (0.110)	0.078 (0.077)	0.239** (0.110)
First-stage F				15485.2				15269.3				15380.7
First-stage F (×Home)				15212.5				14935.5				15040.8
Hausman (p)				0.645				0.195				0.116
N	979,365	917,171	979,365	979,365	979,365	917,171	979,365	979,365	979,365	917,171	979,365	979,365
Outcome: EPA												
Attendance (% obs)												
Attendance	0.006 (0.014)	0.015 (0.016)	-0.008 (0.010)	-0.013 (0.013)	0.017 (0.014)	0.030* (0.016)	0.003 (0.009)	-0.006 (0.012)	0.008 (0.015)	0.023 (0.016)	-0.004 (0.010)	-0.013 (0.013)
Attendance × Home	0.018* (0.009)	0.019 (0.014)	0.018* (0.009)	0.016 (0.014)	-0.004 (0.009)	-0.013 (0.013)	-0.004 (0.009)	0.002 (0.013)	0.008 (0.009)	-0.004 (0.013)	0.008 (0.009)	0.016 (0.014)
First-stage F				15516.2				15295.5				15405.2
First-stage F (×Home)				15253.3				14972.2				15076.1
Hausman (p)				0.448				0.401				0.464
N	976,471	914,450	976,471	976,471	976,471	914,450	976,471	976,471	976,471	914,450	976,471	976,471
Specification												
Strength control	None	None	None	None	Spread	Spread	Spread	Spread	Elo	Elo	Elo	Elo
Venue FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Season FE	Yes	Yes	No	Yes	Yes	Yes	No	Yes	Yes	No	Yes	No
Week FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Game controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Situation controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Instrument	—	—	—	COVID	—	—	—	COVID	—	—	—	COVID

SE in parentheses. *** p<.01, ** p<.05, * p<.10.

B.2.2 Down by Down

Table 18: Offensive success by down (% obs)

	Down 1	Down 2	Down 3
Outcome: First down			
Attendance	0.0115** (0.0056)	0.0066** (0.0033)	0.0094** (0.0044)
Attendance × Home	0.0019 (0.0047)	-0.0037 (0.0030)	-0.0047 (0.0039)
N	444,707	330,018	204,640

Table 18: Offensive success by down (% obs) (*continued*)

	Down 1	Down 2	Down 3
Outcome: Yards gained			
Attendance	-0.0368 (0.1494)	-0.0346 (0.1587)	0.0719 (0.2104)
Attendance $\times$ Home	0.0345 (0.1200)	0.2557** (0.1288)	-0.1196 (0.1751)
N	444,707	330,018	204,640
Outcome: EPA			
Attendance	-0.0120 (0.0162)	0.0065 (0.0218)	0.0546 (0.0396)
Attendance $\times$ Home	0.0082 (0.0126)	0.0240 (0.0175)	-0.0171 (0.0311)
N	443,407	329,039	204,025
Specification			
Strength control	Elo	Elo	Elo
Venue FE	Yes	Yes	Yes
Season FE	Yes	Yes	Yes
Week FE	Yes	Yes	Yes
Game controls	Yes	Yes	Yes
Situation controls	Yes	Yes	Yes
Instrument	—	—	—

Attendance (% obs) SE in parentheses. *** $p < .01$, ** $p < .05$, * $p < .10$.

B.2.3 Spread Attendance 10k

To interpret the coefficients when we use spread as a control we remember that the spread line very well may price in the home team and crowd advantage as argued by Cook (2026). Thus the interacted coefficient of attendance, while still significant, is only 0.023(0.007) and represents the portion of the effect that bookmakers do not account for.

Note that estimating the spread model with only the IV COVID would be invalid as spread contains information about attendance and thus we would be under specified if endogeneity was present. The entire point of the IV model is to deal with potential endogeneity (which we know does not exist) and if that endogeneity does exist then it contaminates the spread variable as well.

Table 19: Attendance and offensive success (play level), spread control

	Spread control			
	OLS	OLS nC	OLS nF	COVID
Outcome: First down				
<i>Attendance (10k)</i>				
Attendance	0.001 (0.000)	0.001** (0.001)	0.000 (0.000)	0.000 (0.001)
Attendance $\times$ Home	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.001 (0.001)
First-stage F				5315.7
First-stage F ($\times$ Home)				4043.1
Hausman (p)				0.554
N	979,365	917,171	979,365	979,365
Outcome: Yards gained				
<i>Attendance (10k)</i>				
Attendance	-0.017 (0.016)	0.010 (0.024)	-0.021* (0.011)	-0.015 (0.018)
Attendance $\times$ Home	0.023*** (0.007)	0.022*** (0.007)	0.023*** (0.007)	0.025 (0.022)

Table 19: Attendance and offensive success (play level), spread control (*continued*)

	OLS	OLS nC	OLS nF	COVID
First-stage F				5315.7
First-stage F ($\times$ Home)				4043.1
Hausman (p)				0.696
N	979,365	917,171	979,365	979,365
Outcome: EPA				
<i>Attendance (10k)</i>				
Attendance	-0.003 (0.002)	0.003 (0.003)	-0.003* (0.002)	-0.001 (0.002)
Attendance $\times$ Home	0.003*** (0.001)	0.003*** (0.001)	0.002*** (0.001)	0.000 (0.003)
First-stage F				5309.3
First-stage F ($\times$ Home)				4040.5
Hausman (p)				0.571
N	976,471	914,450	976,471	976,471
Specification				
Strength control	Spread	Spread	Spread	Spread
Venue FE	Yes	Yes	Yes	Yes
Season FE	Yes	Yes	No	No
Week FE	Yes	Yes	Yes	Yes
Game controls	Yes	Yes	Yes	Yes
Situation controls	Yes	Yes	Yes	Yes
Instrument	—	—	—	COVID

SE in parentheses. *** p<.01, ** p<.05, * p<.10.

B.3 PBP Pen

B.3.1 Pooled

Again we see robustness of our results when estimated on the percentage of observed attendance. Most encouragingly is the Attendance*Home 0.0022(0.0007) coefficient for defensive pass interferences.

Table 20: Penalties (play level, % obs)

	Spread control				Elo control			
	OLS	OLS nC	OLS nF	COVID	OLS	OLS nC	OLS nF	COVID
Outcome: False start								
Attendance	0.0029** (0.0012)	0.0031** (0.0014)	0.0024*** (0.0008)	0.0020* (0.0011)	0.0027** (0.0012)	0.0027** (0.0014)	0.0023*** (0.0008)	0.0021* (0.0011)
Attendance $\times$ Home	-0.0031*** (0.0011)	-0.0035** (0.0015)	-0.0031*** (0.0011)	-0.0030* (0.0015)	-0.0032*** (0.0011)	-0.0036** (0.0015)	-0.0032*** (0.0011)	-0.0032** (0.0015)
First-stage F				15336.6				15446.5
First-stage F ($\times$ Home)				14917.6				15015.3
Hausman (p)				0.817				0.943
N	1,058,548	991,166	1,058,548	1,058,548	1,058,548	991,166	1,058,548	1,058,548
Outcome: Presnap off (ex FS)								
Attendance	0.0004 (0.0006)	0.0008 (0.0007)	0.0005 (0.0004)	0.0003 (0.0005)	0.0005 (0.0006)	0.0008 (0.0007)	0.0006 (0.0004)	0.0003 (0.0005)
Attendance $\times$ Home	-0.0013** (0.0005)	-0.0018*** (0.0007)	-0.0013** (0.0005)	-0.0007 (0.0007)	-0.0014*** (0.0005)	-0.0019*** (0.0007)	-0.0014*** (0.0005)	-0.0008 (0.0007)
First-stage F				15336.6				15446.5
First-stage F ($\times$ Home)				14917.6				15015.3
Hausman (p)				0.495				0.506
N	1,058,548	991,166	1,058,548	1,058,548	1,058,548	991,166	1,058,548	1,058,548
Outcome: Def offside								
Attendance	-0.0018** (0.0008)	-0.0010 (0.0009)	-0.0018*** (0.0005)	-0.0023*** (0.0007)	-0.0016** (0.0008)	-0.0009 (0.0009)	-0.0018*** (0.0005)	-0.0022*** (0.0007)
Attendance $\times$ Home	0.0007 (0.0007)	-0.0004 (0.0010)	0.0007 (0.0007)	0.0016 (0.0010)	0.0006 (0.0007)	-0.0004 (0.0010)	0.0006 (0.0007)	0.0016 (0.0010)
First-stage F				15336.6				15446.5
First-stage F ($\times$ Home)				14917.6				15015.3

Table 20: Penalties (play level, % obs) (*continued*)

	OLS	OLS nC	OLS nF	COVID	OLS	OLS nC	OLS nF	COVID
Hausman (p)				0.243				0.273
N	1,058,548	991,166	1,058,548	1,058,548	1,058,548	991,166	1,058,548	1,058,548
Outcome: Presnap def (ex off)								
Attendance	0.0001 (0.0003)	0.0003 (0.0003)	0.0001 (0.0002)	-0.0001 (0.0003)	0.0001 (0.0003)	0.0003 (0.0003)	0.0001 (0.0002)	-0.0001 (0.0003)
Attendance × Home	-0.0001 (0.0003)	-0.0002 (0.0004)	0.0001 (0.0003)	0.0004 (0.0004)	0.0001 (0.0003)	-0.0002 (0.0004)	0.0001 (0.0003)	0.0004 (0.0004)
First-stage F				15336.6				15446.5
First-stage F (×Home)				14917.6				15015.3
Hausman (p)				0.537				0.548
N	1,058,548	991,166	1,058,548	1,058,548	1,058,548	991,166	1,058,548	1,058,548
Outcome: Off holding								
Attendance	0.0006 (0.0010)	0.0023** (0.0011)	0.0006 (0.0007)	-0.0005 (0.0008)	0.0006 (0.0010)	0.0023** (0.0011)	0.0006 (0.0007)	-0.0005 (0.0008)
Attendance × Home	-0.0012 (0.0009)	-0.0033*** (0.0013)	-0.0012 (0.0009)	0.0009 (0.0012)	-0.0010 (0.0009)	-0.0032** (0.0013)	-0.0010 (0.0009)	0.0010 (0.0012)
First-stage F				15336.6				15446.5
First-stage F (×Home)				14917.6				15015.3
Hausman (p)				0.051				0.055
N	1,058,548	991,166	1,058,548	1,058,548	1,058,548	991,166	1,058,548	1,058,548
Outcome: Def holding								
Attendance	-0.0002 (0.0005)	-0.0004 (0.0005)	-0.0006* (0.0003)	-0.0006 (0.0004)	-0.0003 (0.0005)	-0.0005 (0.0005)	-0.0006** (0.0003)	-0.0006 (0.0004)
Attendance × Home	0.0003 (0.0004)	0.0007 (0.0006)	0.0003 (0.0004)	-0.0000 (0.0006)	0.0003 (0.0004)	0.0007 (0.0006)	0.0003 (0.0004)	-0.0000 (0.0006)
First-stage F				15336.6				15446.5
First-stage F (×Home)				14917.6				15015.3
Hausman (p)				0.434				0.553
N	1,058,548	991,166	1,058,548	1,058,548	1,058,548	991,166	1,058,548	1,058,548
Outcome: Off PI								
Attendance	-0.0003 (0.0003)	0.0000 (0.0004)	-0.0000 (0.0002)	-0.0001 (0.0003)	-0.0003 (0.0003)	-0.0000 (0.0004)	-0.0000 (0.0002)	-0.0001 (0.0003)
Attendance × Home	-0.0001 (0.0003)	-0.0006 (0.0004)	-0.0001 (0.0003)	0.0003 (0.0004)	-0.0001 (0.0003)	-0.0006 (0.0004)	-0.0001 (0.0003)	0.0003 (0.0004)
First-stage F				15336.6				15446.5
First-stage F (×Home)				14917.6				15015.3
Hausman (p)				0.234				0.209
N	1,058,548	991,166	1,058,548	1,058,548	1,058,548	991,166	1,058,548	1,058,548
Outcome: Def PI								
Attendance	-0.0002 (0.0008)	-0.0011 (0.0009)	-0.0016*** (0.0006)	-0.0018** (0.0007)	-0.0003 (0.0008)	-0.0012 (0.0009)	-0.0017*** (0.0006)	-0.0019** (0.0007)
Attendance × Home	0.0022*** (0.0007)	0.0032*** (0.0010)	0.0022*** (0.0007)	0.0010 (0.0010)	0.0023*** (0.0007)	0.0033*** (0.0010)	0.0023*** (0.0007)	0.0011 (0.0010)
First-stage F				15336.6				15446.5
First-stage F (×Home)				14917.6				15015.3
Hausman (p)				0.008				0.010
N	1,058,548	991,166	1,058,548	1,058,548	1,058,548	991,166	1,058,548	1,058,548
Specification								
Strength control	Spread	Spread	Spread	Spread	Elo	Elo	Elo	Elo
Venue FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Season FE	Yes	Yes	No	No	Yes	Yes	No	No
Week FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Game controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Situation controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Instrument	—	—	—	COVID	—	—	—	COVID

Attendance (% obs). SE in parentheses. *** p<.01, ** p<.05, * p<.10.

B.3.2 Down by Down

Table 21: Penalties by down (% obs)

	Down 1	Down 2	Down 3
Outcome: False start			
Attendance	0.0015 (0.0016)	0.0025 (0.0020)	0.0055* (0.0029)
Attendance × Home	-0.0023* (0.0014)	-0.0034** (0.0017)	-0.0048** (0.0024)
N	480,476	353,511	224,561
Outcome: Presnap off (ex FS)			

Table 21: Penalties by down (% obs) (*continued*)

	Down 1	Down 2	Down 3
Attendance	0.0009 (0.0009)	-0.0004 (0.0009)	0.0010 (0.0015)
Attendance × Home	-0.0016** (0.0008)	-0.0006 (0.0007)	-0.0021 (0.0013)
N	480,476	353,511	224,561
Outcome: Def offside			
Attendance	-0.0013 (0.0010)	-0.0018 (0.0011)	-0.0019 (0.0020)
Attendance × Home	-0.0001 (0.0009)	0.0005 (0.0010)	0.0024 (0.0018)
N	480,476	353,511	224,561
Outcome: Presnap def (ex off)			
Attendance	0.0001 (0.0005)	0.0000 (0.0005)	0.0001 (0.0007)
Attendance × Home	0.0001 (0.0004)	-0.0004 (0.0004)	0.0009 (0.0006)
N	480,476	353,511	224,561
Outcome: Off holding			
Attendance	0.0019 (0.0016)	-0.0014 (0.0016)	0.0008 (0.0016)
Attendance × Home	-0.0026* (0.0014)	0.0007 (0.0015)	-0.0004 (0.0015)
N	480,476	353,511	224,561
Outcome: Def holding			
Attendance	-0.0010 (0.0007)	-0.0002 (0.0008)	0.0009 (0.0012)
Attendance × Home	0.0004 (0.0006)	0.0011* (0.0007)	-0.0011 (0.0010)
N	480,476	353,511	224,561
Outcome: Off PI			
Attendance	-0.0004 (0.0004)	-0.0002 (0.0005)	-0.0004 (0.0007)
Attendance × Home	-0.0001 (0.0004)	-0.0002 (0.0005)	0.0000 (0.0007)
N	480,476	353,511	224,561
Outcome: Def PI			
Attendance	0.0018* (0.0011)	-0.0033** (0.0014)	0.0000 (0.0020)
Attendance × Home	0.0012 (0.0010)	0.0030** (0.0012)	0.0033* (0.0017)
N	480,476	353,511	224,561
Specification			
Strength control	Elo	Elo	Elo
Venue FE	Yes	Yes	Yes
Season FE	Yes	Yes	Yes
Week FE	Yes	Yes	Yes
Game controls	Yes	Yes	Yes
Situation controls	Yes	Yes	Yes
Instrument	—	—	—

Attendance (% obs) SE in parentheses. *** p<.01, ** p<.05, * p<.10.